D.C.MACHINES

A machine which works on direct current is defined as a **D.C.Machine.** D C machines are electrical machines which deal with conversion of one form of energy to another. A machine which converts mechanical energy to electrical energy is called a dc generator, while a dc machine which converts electrical into mechanical energy is known as dc motor.

3A -DC GENERATOR

Construction of DC Machines

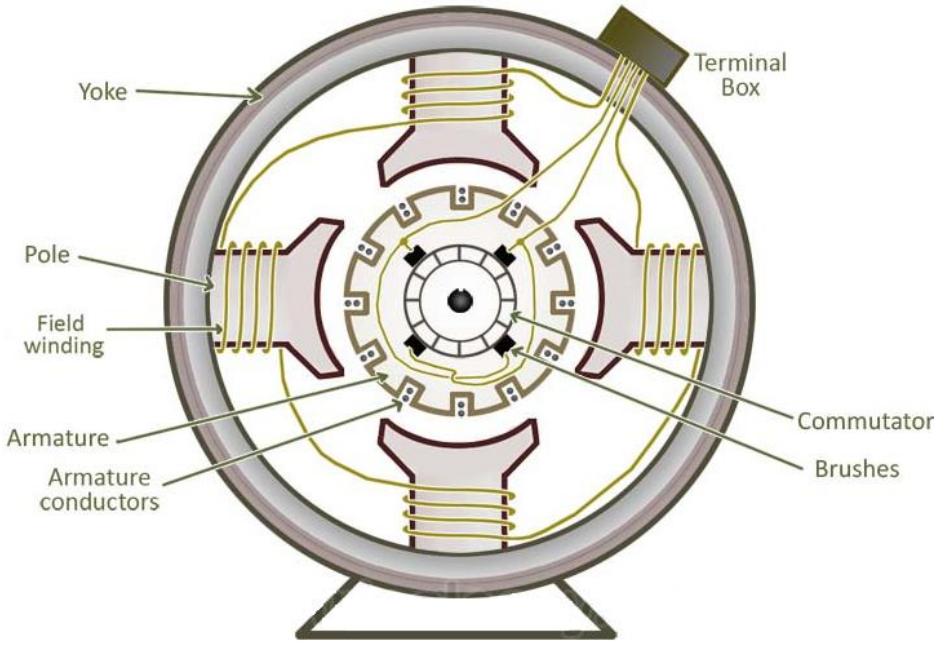


Fig.3.1

The construction of a DC machine is as shown in the figure.3.1

1. **Yoke**

Functions: It serves the purpose of outermost cover of the d.c. machine, so that insulating materials get protected from harmful atmospheric elements like moisture, dust and various gases like $S0_2$, acidic fumes etc.

- It provides mechanical support to the poles.
- It forms a part of the magnetic circuit. It provides a path of low reluctance for magnetic flux.

Choice of material:

- It is prepared by using cast iron because it is Cheapest and provides low reluctance path. For large machines rolled steel, cast steel, Silicon steel is used which provides high permeability.

2. Poles

Each pole is divided into two parts namely,

i) Pole core and ii) Pole shoe

Functions:

- Pole core basically carries a field pole winding which is necessary to produce core flux.
- Pole shoe directs the flux produced through air gap to armature core, to the next pole.
- Pole shoe enlarges the area of the armature core to come across the flux, which is necessary to produce larger induced e.m.f to achieve this, pole shoe has been given a particular shape.

Choice of material:

- It is made up of magnetic material like cast iron or cast steel. The laminations of required size and shape are stamped together to get a pole which is then bolted to the yoke.

3. Field Winding

The field winding is wound on the pole core with a definite direction.

Functions:

- To carry current due to which pole core, on which the field winding is placed, behaves as an electromagnet, producing necessary flux.
- As it helps in producing the magnetic field i.e. exciting the pole as an electromagnet it is called Field winding or Exciting winding.
- Field winding is wound in such a direction that alternate 'N' and ’S’ poles are formed. The total number of poles is denoted as P.

Choice of material:

- Copper is the best material as it is a good conductor and has good pliability.

4. Armature

The armature is further divided into two parts namely,

i) Armature core and ii) Armature winding

i) Armature core

Functions:

- Armature core provides house for armature winding.
- Armature core is cylindrical in shape mounted on the shaft. It consists of slots on its periphery and the air ducts to permit the air flow through armature which serves cooling purpose.
- Laminated construction is used to keep the eddy current losses to a minimum value.

Choice of material:

- To provide a path of low reluctance to the magnetic flux produced by the field winding.

ii) Armature winding

Functions:

- Armature conductors are interconnected to form armature winding and are placed in the slots provided on the armature core periphery.
- When the armature is rotated, in case of generator, magnetic flux gets cut by armature conductors and e.m.f. gets induced in Shaft .

Choice of material:

- As armature winding carries entire current it is to be made up of conducting material, which is copper.

The construction of armature winding of DC machine can be of two types:

- Lap Winding: In this case, the number of parallel paths between conductors (A) is equal to the number of poles (P), i.e. A = P.
- Wave Winding: In this case, the number of parallel paths between conductors (A) is always equal to 2 irrespective of the number of poles, i.e. A = 2.

5. Commutator

Functions:

- The basic nature of e.m.f. induced in the armature conductors is alternating. This needs rectification in case of d.c. generator, which is done using a device called Commutator.
- Commutator collects the current from armature conductor and converts the developed alternating emf to unidirectional emf.
- In case of d.c. motors, it produces unidirectional torque.

Choice of material:

- It is cylindrical in shape and is made up of wedge-shaped segments of high conductivity copper. These segments are insulated from each other by a thin layer of mica.

6. Brushes

Functions:

- Brushes are stationary and resting on the surface of the Commutator.
- It collects current from commutator and makes it available to the stationary external circuit.

Choice of material:

- Brushes are normally made up of soft material like Carbon to avoid wear and tear of Commutator.

Shaft: Shaft of DC machine is coupled to the load to transfer mechanical power. If the machine is used as a DC motor, mechanical power is transferred from the motor to load. If the machine is used as a DC generator, mechanical power is transferred from prime mover to the machine. Armature core, bearing, commutator etc. are mounted on the Shaft.

Working Principle

All generators work based on the principle of Faraday's laws of electromagnetic induction. The generating action requires following basic components to exist

a) Conductor or coil
b) Magnetic field
c) Relative motion between conductor and flux

Large numbers of conductors are connected together in specific manner to form a winding. It is called as armature winding and the part on which this winding is kept is called armature of DC machine. The necessary magnetic field is produced by current carrying conductor which is called as field winding.

Conductors placed on the armature are rotated with the help of external device called as prime movers. Commonly used prime movers are diesel engine, steam engine, water engine etc.

An emf induced due to physical movement of the coil with respect to flux is called dynamically induced emf whose direction is obtained using Fleming's Right hand rule.

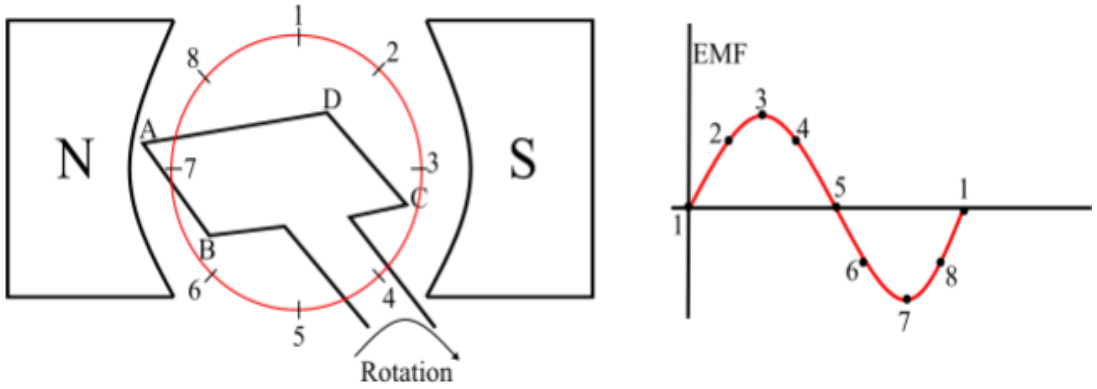
Induced emf is given by,

$$e = Blv \sin\backslash \text{ volts}$$

From this equation the basic nature of induced emf in DC generator is purely sinusoidal. To get DC voltage, a device called Commutator is used to convert the alternating emf to unidirectional emf.

Working Principle:

Working of a DC Generator Consider a single loop DC generator (as shown in the figure), in this a single turn loop 'ABCD' is rotating clockwise in a uniform magnetic field with a constant speed. When the loop rotates, the magnetic flux linking the coil sides 'AB' and 'CD' changes continuously. This change in flux linkage induces an EMF in coil sides and the induced EMF in one coil side adds the induced EMF in the other.



The EMF induced in a DC generator can be explained as follows

- When the loop is in position-1, the generated EMF is zero because, the movement of coil sides is parallel to the magnetic flux.
- When the loop is in position-2, the coil sides are moving at an angle to the magnetic flux and hence, a small EMF is generated.
- When the loop is in position-3, the coil sides are moving at right angle to the magnetic flux, therefore the generated EMF is maximum.
- When the loop is in position-4, the coil sides are cutting the magnetic flux at an angle, thus a reduced EMF is generated in the coil sides.
- When the loop is in position-5, no flux linkage with the coil side and are moving parallel to the magnetic flux. Therefore, no EMF is generated in the coil.
- At the position-6, the coil sides move under a pole of opposite polarity and hence the polarity of generated EMF is reversed. The maximum EMF will generate in this direction at position-7 and zero when at position-1. This cycle repeats with revolution of the coil.

It is clear that the generated EMF in the loop is alternating one. It is because any coil side (say AB) has EMF in one direction when under the influence of N-pole and in the other direction when under the influence of S-pole. Hence, when a load is connected across the terminals of the generator, an alternating current will flow through it. Now, by using a commutator, this alternating emf generated in the loop can be converted into direct voltage. We then have a DC generator.

Emf Equation of Dc Generator

Let P - No. of poles.

ϕ- flux produced by each pole (in wb)

N- speed of armature (RPM)

Z- Total no of armature conductor.

A - No. of parallel paths

Emf gets induced in the conductor according to Faraday's laws of electromagnetic induction.

Hence average value of induced emf in each conductor is e = $\frac{d\phi}{dt}$ ……. (1)

Consider one revolution of conductor, flux cut by conductor in 1 revolution is $d\phi = \phi x P$.....(2)

Time required to complete 1 revolution is

$dt = \frac{60}{N}$ sec …… (3)

Substituting eq.(3) & (2) in eq. (1)

$e = \frac{\frac{\phi * P}{60}}{\frac{}{N}} = \frac{\phi * P * N}{60}$.

There are totally Z armature conductors with A parallel paths. Hence $\left(\frac{Z}{A}\right)$ No. of conductors are in series and emf remains same across all parallel paths.

The total emf induced = emf induced per conductor * No. of conductors / parallel path.

$$E_g = \frac{\phi * P * N}{60} \text{ X } \left(\frac{Z}{A}\right)$$

$$E_g = \frac{\phi * P * N * Z}{60 * A} \text{ volts.}$$

For lap winding, A= P

$$E_g = \frac{\phi * N * Z}{60} \text{ volts}$$

For wave winding, A= 2

$$E_g = \frac{\phi * P * N * Z}{120} \text{ volts.}$$

Problems:

1.1 A 4 pole, 1500rpm DC generator has lap wound armature conductors having 24 slots with 10 conductors per slot. If the flux per pole is 0.04 wb. Calculate the emf generated in the armature, what would be the emf if the winding is wave connected.

Solution:

$$E_g = \frac{\phi * P * N * Z}{60 * A} \text{ volts.}$$

Z = No. of slots * Cond./slot

24*10 = 240

i. For Lap Winding; A= P = 4

$$E_g = \frac{0.04*240*1500*4}{60*4} = \textbf{240 volts}$$

ii. For Wave Winding ; A= 2

$$E_g = \frac{0.04*240*1500*4}{60*2} = \textbf{480 volts.}$$

0.3 Problems on DC Generator

Model paper 1 8 c) A 4 pole lap wound DC shunt generator delivers 200 A at terminal voltage of 250 V. It has a field and armature resistance of 50 Ω and 0.05 Ω respectively. Neglecting brush drop determine i) armature current ii) current per parallel path iii) EMF generated iv) power delivered.

Solution:

i) armature current

$$I_{sh} = \frac{V}{R_{sh}} = \frac{250}{50}$$
$$= 5A$$

$$I_a = I_L + I_{sh} = 200 + 5$$
$$= 205A$$

ii) current per parallel path

$$= \frac{I_a}{4} = \frac{205}{4}$$
$$= 51.25A$$

iii) EMF generated

$$E_g = V + I_a R_a = 250 + 205 \times 0.05$$
$$= 260.25V$$

iv) power delivered

$$= E_g I_a = 260.25 \times 205$$
$$= 53.35kW$$

Model paper 2 8 c) A 30 kW 300 V DC shunt generator has a armature and field resistance of 0.05 Ω and 100 Ω respectively. Calculate the power delivered by the armature when it delivers full output power.

Solution:

$$I_L = \frac{P}{V} = \frac{30 \times 10^3}{300}$$
$$= 100A$$

$$I_{sh} = \frac{V}{R_{sh}} = \frac{300}{100}$$
$$= 3A$$

$$Ia = I_L + I_{sh} = 100 + 3$$
$$= 103A$$

$$E_g = V + I_a R_a = 300 + 103 \times 0.05$$
$$= 305.15V$$

power delivered

$$= E_g I_a = 305.15 \times 103$$
$$= 31.43kW$$

Model paper 3 8 c) A 4 pole generator with wave wound armature has 51 slots each having 24 conductors rotate to give an induced emf of 220 V. What will be the voltage developed if the winding is lap and the armature rotates at the same speed

Solution:

$$\begin{aligned} E_g &= \frac{\phi Z N}{60} \times \frac{P}{A} \\ 220 &= \frac{0.01 \times 51 \times 24 \times N}{60} \times \frac{4}{2} \\ N &= 539.2156rpm \end{aligned}$$

Voltage developed with lap wound

$$\begin{aligned} E_g &= \frac{\phi Z N}{60} \times \frac{P}{A} \\ &= \frac{0.01 \times 51 \times 24 \times 539.215}{60} \times \frac{4}{4} \\ &= 110V \end{aligned}$$

2020 Jan 7 c) A shunt wound DC generator delivers 496 A at 440 V to load. The resistance of the shunt field coil is 110 Ω and that of armature winding is 0.02 ohm. Calculate the emf induced in the coil armature.

Solution:

$$\begin{aligned} E &= V - I_a R_a \\ &= 440 - 496 \times 0.02 \\ &= 440 - 9.92 \\ &= 430.08V \end{aligned}$$

Load at which the maximum efficiency is

$$\begin{aligned} I_a &= I_L + I_{sh} \\ &= 50\sqrt{\frac{400}{600}} \end{aligned}$$

2019 June 7 b) A 4 pole 230 V DC series wave connected armature with 1254 conductors with flux/pole of 22 mWb, takes 50A for motoring. The

armature and series filed coil resistances are 0.3Ω and 0.2Ω respectively. Calculate the speed and torque developed in Watts.

Solution:

i) For lap wound connection. The emf generated in the armature is

$$\begin{aligned} E &= V + I_a(R_a + R_{sc}) \\ &= 230 + 50(0.3 + 0.2) \\ &= 255V \end{aligned}$$

$$\begin{aligned} E &= \frac{\phi ZN}{60} \times \frac{P}{A} \\ 255 &= \frac{22 \times 10^{-3} \times 1254 \times N}{60} \times \frac{4}{2} \\ N &= 277rpm \end{aligned}$$

$$\begin{aligned} T_a &= 0.159 \times \phi I_a Z \times \frac{P}{A} \\ &= 0.159 \times 22 \times 10^{-3} \times 50 \times 1254 \times \frac{4}{2} \\ &= 438.6Nm \end{aligned}$$

2019 June 8 b) A shunt generator has 4 pole with lap wound armature having 24 slots with 10 conductors/slot. If the flux/pole is 0.04 Wb and the speed is 1500 rpm, calculate the emf generated in the armature. What would be generated emf if the winding is wave connected?

Solution:

i) For lap wound connection. The emf generated in the armature is

$$\begin{aligned} E &= \frac{\phi ZNP}{60A} \\ &= \frac{0.04 \times (24 \times 10) \times 1500 \times 4}{60 \times 4} \\ &= 240 \end{aligned}$$

ii) For wave wound connection. The emf generated in the armature is

$$\begin{aligned} E &= \frac{\phi ZNP}{120} \\ &= \frac{0.04 \times (24 \times 10) \times 1500 \times 4}{120} \\ &= 480 \end{aligned}$$

2019 Jan 7 b) A 4 pole armature of dc generator has 624 lap connected conductors and is driven at 1200 rpm. Calculate the useful flux per pole required to generate an emf of 250 V.

Solution:

$$\begin{aligned} E &= \frac{\phi ZN}{60} \times \frac{P}{A} \\ \phi &= \frac{E60}{ZN}\frac{A}{P} \\ &= \frac{250 \times 60 \times 4}{624 \times 1200 \times 4} \\ &= 0.02wb \end{aligned}$$

2019 Jan 8 c) A shunt generator running at 500 rpm delivers 50 kW at 200 V. The armature and field resistances are 0.02 Ω and 40 Ω respectively. Calculate generated emf if brush drop of 1 V per brush.

Solution:

$$\begin{aligned} I_{sh} &= \frac{V}{R_{sh}} = \frac{200}{40} \\ &= 5A \\ I_L &= \frac{P}{V} = \frac{50 \times 10^3}{200} \\ &= 250A \\ I_a &= I_L + I_{sh} = 250 + 5 \\ &= 255A \end{aligned}$$

$$\begin{aligned} E &= V + I_a R_a + Brush\ drop \\ &= 200 + 255 \times 0.02 + 2 \\ &= 207.1V \end{aligned}$$

2019 Jan 4 c (17 scheme)) A 4 pole DC shunt generator with lap connected armature has field and armature resistances are 50 Ω and 0.1 Ω respectively. If the generator supplies sixty 100 V 40 W lamps calculate the total armature current, the current in each armature conductor and the generated EMF. Take 1 V per brush as contact drop.

Solution:

Power delivered is

$$\begin{aligned} P &= 60 \times 40 \\ &= 2400W \end{aligned}$$

$$\begin{aligned} I_{sh} &= \frac{V}{R_{sh}} = \frac{100}{50} \\ &= 2A \\ I_L &= \frac{P}{V} = \frac{2400}{100} \\ &= 24A \\ I_a &= I_L + I_{sh} = 24 + 2 \\ &= 26A \end{aligned}$$

Current in each conductor is

$$\frac{I_a}{A} = \frac{26}{4}$$
$$= 6.5A$$

$$E_g = V + I_a R_a + Brush\ drop$$
$$= 100 + 26 \times 0.1 + 2$$
$$= 104.6V$$

2019 Dec 4 b (17 scheme)) An 8 pole lap connected armature has 960 conductors, a flux of 40 mWb per pole and a speed of 400 rpm. Calculate the emf generated. If the armature were wave connected at what speed must it be driven to generate 400 V.

Solution:

$$E_g = \frac{\phi ZN}{60} \times \frac{P}{A}$$
$$= \frac{40 \times 10^{-3} \times 960 \times 400}{60} \times \frac{8}{8}$$
$$= 256$$

When wave connected

$$E_g = \frac{\phi ZNP}{120}$$
$$N = \frac{120 \times E_g}{\phi ZP}$$
$$N = \frac{120 \times 256}{40 \times 10^{-3} \times 960 \times 8}$$
$$= 100\ rpm$$

2019 Dec 4 b (17 scheme)) An 8 pole DC generator has 500 armature conductors, and useful flux/ple of 0.065Wb. What will be the emf generated if it is lap connected and runs at 1000 rpm. At what speed it must be driven to produce the same emf if it is wave connected.

Solution:

When lap connected:

$$E_g = \frac{\phi ZNP}{60A}$$
$$= \frac{0.065 \times 500 \times 1000 \times 8}{60 \times 8}$$
$$= 541.67$$

When wave connected

$$E_g = \frac{\phi ZNP}{120}$$
$$N = \frac{120 \times E_g}{\phi ZP}$$
$$N = \frac{120 \times 541.67}{0.065 \times 500 \times 8}$$
$$= 250\ rpm$$

2018 Dec 4 c (15 scheme)) An 8 pole lap connected armature has 40 slots with 12 conductors per slot, generate a voltage of 500 V. Determine the speed at which it is running if the flux pole is 50 mWb.

Solution:

When lap connected:

$$E_g = \frac{\phi ZNP}{60A}$$
$$N = \frac{60 \times A \times E_g}{\phi ZP}$$
$$= \frac{60 \times 8 \times 500}{50 \times 10^{-3} \times 40 \times 12 \times 8}$$
$$= 1250\ rpm$$

2019 Dec 4 c (15 scheme)) A 4 pole 1500 rpm DC generator has lap wound armature having 24 slots with 10 conductors per slot. If the flux per pole is 0.04 wb. Calculate the emf generated in the armature. What would be the generated emf if the winding is wave connected.

Solution:

When lap connected:

$$E_g = \frac{\phi ZNP}{60A}$$
$$= \frac{0.04 \times 24 \times 10 \times 1500 \times 4}{60 \times 4}$$
$$= 240V$$

When wave connected

$$E_g = \frac{\phi ZNP}{120}$$
$$= \frac{0.04 \times 24 \times 10 \times 1500 \times 4}{60 \times 2}$$
$$= 480V$$

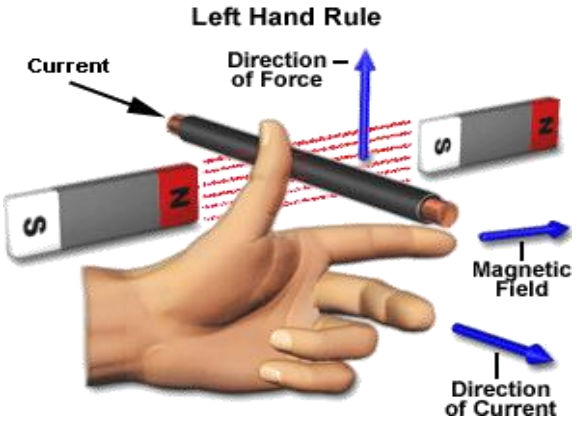
DC MOTOR

A DC motor is a rotating machine which converts DC electrical energy into mechanical energy.

1.1 Working Principle of DC Motor

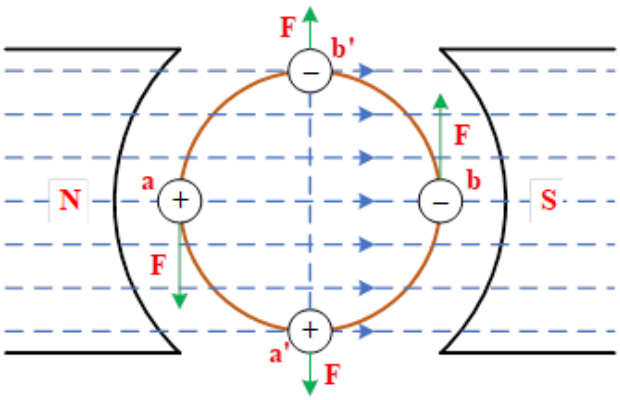
A DC motor is an electromagnetic device, which converts electrical power into mechanical power. DC motors operate on Faraday's principle of electromagnetism, which states that "when a current carrying conductor is placed in varying magnetic field or a conductor is moved in stationary magnetic field, it experiences a mechanical force". The direction of this force, given by Fleming's left-hand rule, is always in perpendicular to current and magnetic field; and its magnitude is given by $F = BIL$. Where, F is force, B is magnetic field, I is current, and L is length of conductor within the magnetic field.

Fleming's left-hand rule: If we stretch the index finger, middle finger and thumb of our left hand to be perpendicular to each other in such a way that index finger is in the direction of magnetic field and middle finger is in the direction of current, then thumb indicates the direction of mechanical force experienced by the current carrying conductor.



Consider that the armature has only two coils a and b, which are placed between two field magnets as shown below in the figure. When the terminals of the motor are connected to DC supply:

i) The field magnets are excited and develop alternate north (N) and south (S) poles with the direction of magnetic field from N to S;
ii) The armature coils carry currents. All coils under N-pole carry currents in one direction (inwards, represented by crosses) while all the conductors under S-pole carry currents in the opposite direction (outwards, represented by dots).



According to Fleming's left-hand rule, the coil a experiences force F in the downward direction and the coil b experiences force F in the upward direction. These forces tend to rotate the armature in anti-clockwise direction.

Now, let the armature rotate by 90° in anti-clockwise direction and the coils a and b occupy positions a' and b' respectively. In this position, they experience a force F in the opposite directions along the same line and the resultant force developed by them is zero.

If the armature contains only these two coils, then the armature would stop in the position $a'b'$. But the armature consists of several uniformly distributed coils in the slots of armature, which are connected together and experiencing force to rotate the armature in anti-clockwise direction.

Also, for the armature to experience a continuous anti-clockwise force, it is necessary to reverse the current directions of coils a and b after they cross the positions a' and b' respectively. Otherwise, the armature experiences a pulsating force in the position $a'b'$. This reversal of current direction can be done by the commutator and thus the direction of force acting on

armature remains same in the anti-clockwise direction, which continuously rotates the armature.

Back Emf

When conductor cuts magnetic lines of force, emf gets induced in conductor which is called generating action. In DC motor after motoring action armature starts rotating and armature conductors cuts the main flux. There exists generating action in a motor after every motoring action.

According to lenz's law induced emf always opposes the cause producing it. In DC motor supply voltage is because for armature current and in turn motoring action. Hence this induced emf opposes the supply voltage. As this emf opposes the supply voltage it is called back emf and is denoted by E_b

$$E_b = \frac{\phi * P * N * Z}{60 * A} \text{ volts}$$

Significance of Back Emf

For constant supply voltage,

$E_b \alpha N$

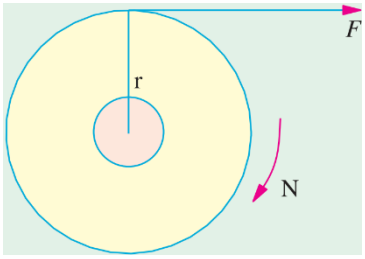
Where N is the speed of armature in RPM

When load is high, motor tries to slow down. So the speed of motor reduces due to which back emf also reduces. So the net voltage across the armature ($V-E_b$) increases and motor draws more armature current and motor draw more armature current. Hence the torque on armature increases.

Similarly when load on motor is decreased speed of motor increases and also back emf. This causes the net voltage ($V-E_b$) to reduce which eventually reduces current drawn by armature which is just enough to produce the torque required to drive the load. So, back emf regulates the flow of armature current making DC motor a self regulating machine. This is the practical significance of back emf.

1.2 Torque Equation

Torque is the turning or twisting moment of a force about an axis. It is measured by the product of force and radius at which this force acts.



Consider the armature of DC motor have radius r and F is the force acting tangential to its surface, which causes it to rotate at speed N as shown in figure. Then the torque developed by the force F on armature is given by,

$$T_a = F \times r \text{ (in Nm)}$$

Work done by this force in one revolution = Force × Distance covered in one revolution $= F \times 2\pi r$ (in Ws).

Power developed by the armature (P_d) = Work done in one second $= F \times 2\pi r \times$ Number of revolutions per second.

$$P_d = F \times 2\pi r \frac{N}{60} = \frac{2\pi N}{60}(F \times r) = \frac{2\pi N}{60} T_a \text{ (in W)}$$

As electrical equivalent of the mechanical power developed (P_d) by armature of DC motor is equal to $E_b I_a$. Thus,

$$P_d = \frac{2\pi N}{60} T_a = E_b I_a = \frac{\phi Z P N}{60A} I_a$$

$$\Rightarrow T_a = \frac{\phi Z I_a}{2\pi}\left(\frac{P}{A}\right) = 0.159 \phi Z I_a \left(\frac{P}{A}\right)$$

The above equation gives the total torque developed by armature, which includes iron losses and mechanical losses of the motor. The actual torque available at the shaft is known as shaft torque (T_{sh}), which is less than the torque lost by iron losses and mechanical losses of the motor. Therefore,

$$T_{sh} = T_a - T_L$$

Where, $T_{sh} =$ Shaft torque, $T_a =$ Armature torque, $T_L =$ torque lost by iron losses and mechanical losses.

$$T_{sh} = \frac{\text{Output power of the motor } (P_{sh}) \times 60}{2\pi N}$$

1.3 Types of DC Motor

Depending on the way in which field winding are connected to the armature, DC motors are classified into three types: (1) DC Shunt Motor (2) DC Series Motor (3) DC Compound Motor.

1.3.1 DC Shunt Motor

In DC shunt motor, the field winding is connected in parallel (shunt) with armature as shown in the figure. Hence, the full voltage is applied across the field winding. Shunt winding is made with large number of turns and has high resistance. It takes only small current which is less than 5% of the rated armature current.

Due to the applied voltage V, line current (I_L) flows through the line, shunt field current (I_{sh}) flows through the shunt field winding and armature current (I_a) flows through the armature conductors, such that $I_L = I_a + I_{sh}$

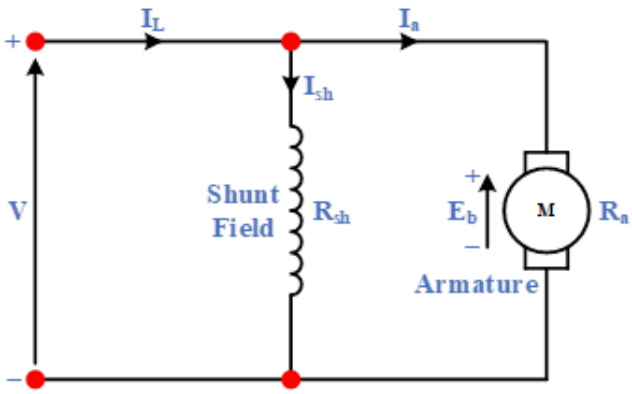
The shunt field current is given by, $I_{sh} = V/R_{sh}$

The armature current is given by, $I_a = I_L - I_{sh}$

The voltage equation of DC shunt motor is given by,

$$V = E_b + I_a R_a + V_b$$

Where, V_b is the brush voltage drop.



Now, multiplying both sides by I_a, we get

$$VI_a = E_b I_a + I_a^2 R_a + V_b I_a$$

Where, VI_a = Electrical input to the armature

$E_b I_a$ = Electrical equivalent of mechanical power developed in the armature

$I_a^2 R_a$ = Copper loss in the armature

$V_b I_a$ = Power loss in brush contact

1.3.2 DC Series Motor

In DC series motor, field winding is connected in series with the armature as shown in figure. Therefore, the field winding carries whole of the load current (armature current). That is why series winding is designed with few turns of thick wire and has low resistance.

Due to the applied voltage V, line current (I_L) flows through the line, series field winding and armature conductors, such that $I_L = I_{se} = I_a$.

The voltage equation of DC series motor is given by,

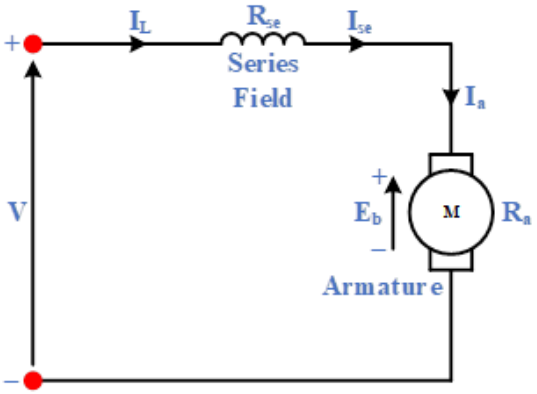
$$V = E_b + I_a(R_a + R_{se}) + V_b$$

Where, V_b is the brush voltage drop.

Now, multiplying both sides by I_a, we get

$$VI_a = E_b I_a + I_a^2(R_a + R_{se})$$

Where, VI_a = Electrical input to the armature



$E_b I_a$ = Electrical equivalent of mechanical power developed in the armature

$I_a^2 R_a$ = Copper loss in the armature

$I_a^2 R_{se}$ = Copper loss in the field

$V_b I_a$ = Power loss in brush contact

1.4 Applications of DC Motors

Type of DC Motor	Characteristics	Applications	
Shunt	• Approximately constant speed from no load to full load • Medium starting torque	• Lathe machines • Machine tools • Blowers and fans • Drilling machines	• Centrifugal pumps • Reciprocating pumps • Boring machines • Spinning and weaving machines
Series	• Variable speed from no load to full load • High Starting torque	• Trolleys • Cranes • Hoists • Conveyors • Hair driers	• Electric traction work • Electric locomotives • Vacuum Cleaners • Sewing machines • Air compressors

1.1 A 500V shunt motor has 4 poles and wave connected winding with 492 conductors. The flux per pole is 0.05Wb. The full load current is 20Amp, armature and shunt field winding resistances are 0.1∧ and 250∧ respectively as shown in the fig.423. Calculate the speed and torque developed.

Solution:

$I_L = I_{sh} + I_a$

$I_{sh} = \frac{V}{R_{sh}} = \frac{500}{250} =$ **2 Amp**

$I_a = I_L - I_{sh}$ ---------- (1)

$I_a = 20 - 2 =$ **18A**

$V - I_aR_a - V_{brush} - E_b = 0$

$V_{brush} = 0$

$E_b = 500 - (18*0.1) =$ **498.2 Volts**

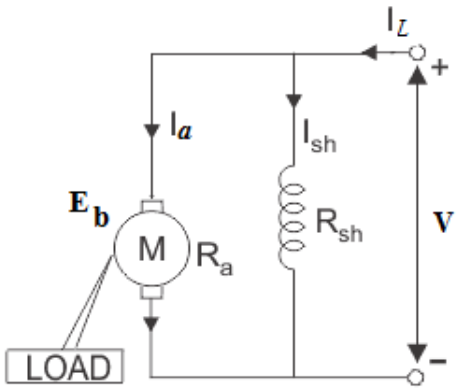


fig.4.23

$$E_b = \frac{\phi * P * N * Z}{60 * A} \text{ volts}$$

$$N = \frac{E_b * 60 * A}{* Z * P} = \mathbf{607.56\ rpm}$$

$$T_a = \frac{0.159\ Ia\ PZ}{A} \text{ N-m}$$

$$T_a = \frac{0.159 * 0.05 * 18 * 2 * 492}{2} \text{ N-m}$$

T_a = 140.81 N-m

4.3 A 4 pole, 220 V. lap connected DC shunt motor has 36 slots, each slot containing 16 conductors. It draws a current of 40 A from the supply. The field resistance and armature resistance are 110, 0.1 respectively. The motor develops an output power of 6 kW. The flux per pole is 40 mwb. Calculate

a) the speed b) the torque developed by the armature and c) the shaft torque

Sol.:

P = 4,

V = 220 volts,

slots = 36,

A = P = 4

Conductors/slot = 16,

Z = 36 × 16 = 576

I_L = 40 A

Rsh = 110

R_a= 0.1

P output = 6 KW, ∅ = 40×10^{-3} wb

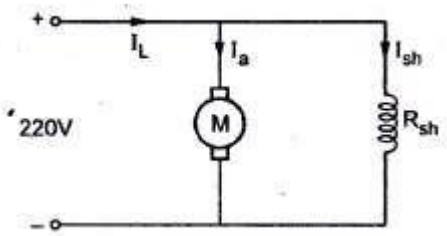


Fig.4.24

Ish = $\frac{V}{R_{sh}}$ = 2A

IL = Ia+Ish

Ia = IL - Ish = 40 - 2 = 38A

Eb = V - IaRa

Eb = 220 - (38) (0.1) = **216.2 volts**

Now, Eb =$\frac{\emptyset ZNP}{60A}$;

N = $\frac{60*A*E_b}{\emptyset*Z*P}$ = **563 rpm**

Ta = 0.159 ∅ Ia (ZP/A) = **139.20 N-m**

P $_{out}$ = T $_{sh}$ * ($\frac{2\pi N}{60}$)

Tsh = = 101.76 N-m

T sh = 101.76 N-m

4.4 A 220*Volts*, 8.8 Kw, 1000rpm, DC shunt generator has Ra of 0.5∧ and field resistance of 110∧ . The brush drop is 1 V/brush. Find the armature Voltage at full load?

Sol :

$V_t = 220V$

$R_{sh} = 110\Omega$

$P_t = 8.8kw$

$P_t = V_t x I_L$

$I_L = \frac{P_t}{V_t} = \frac{8.8x10^3}{220} = 40A$

$I_{sh} = \frac{V_t}{R_{sh}} = \frac{220}{110} = 2A$

$I_a = I_L + I_{sh}$

$= 40 + 2 \Rightarrow 42A$

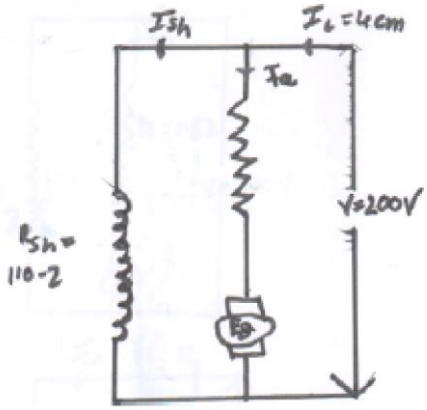


Fig.4.26

Generated Voltage

$E_g - I_a R_a - V_t - 2V_B = 0$

$E_g = I_a R_a + V_t + 2V_B$

$= 42x0.5x220 + 2(1)$

$E_g = 243$ Volts

Problems on DC Motor

P1] A 500V, dc shunt motor has 4 poles, and wave connected armature winding with 492 conductors. The flux per pole is 0.05 Weber. The full load current is 20A. The armature and shunt field resistances are respectively 0.1Ω & 250Ω. Calculate the speed and gross developed torque

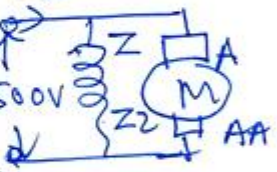
Data given

$V = 500$ volts, $P = 4$

$A = 2$ (wave), $Z = 492$

$\phi = 0.05$ wb, $I_L = 20A$

$R_a = 0.1\Omega$ $R_f = 250\Omega$



$$I_f = \frac{V}{R_f} = \frac{500}{250} = 2A$$

$$I_a = I_L - I_f = 20 - 2 = 18A$$

(I) $$N = \frac{(V - I_a R_a)\,60A}{\phi Z P} = \frac{[500 - 18 \times 0.1]\,60 \times 2}{0.05 \times 492 \times 4}$$

$$\boxed{N = 607.56 \text{ rpm}}$$

(II) $$T_a = 0.159\,\phi Z P\, I_a/A$$

$$= 0.159 \times 0.05 \times 492 \times 4 \times 18/2$$

$$\boxed{T_a = 140.81 \text{ N-M}}$$

P2] Find the useful flux per pole on no load of a 250V, 6 pole, dc shunt motor having two circuit connected armature winding (or wave connected) with 110 turns. At normal working temperature, the overall armature resistance including brushes is 0.2Ω. The armature current is 13.3A at the no load speed of 908 rpm.

Data given

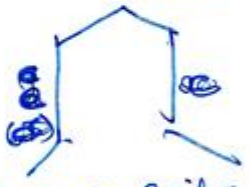
$V = 250$ Volts, $P = 6$, $A = 2$, $Z = 2C = 2T$

$Z = 2 \times 110 = 220$

$Z = \underline{220}$

$I_a = 13.3A$, $N = 908$ rpm

$R_b + R_a = 0.2\Omega$



1 Turn coil = 2 conductors

$E_b = V - I_a R_a - V_b$

$= V - I_a R_a - I_a R_b$

$= V - I_a (R_a + R_b)$

$E_b = 250 - 13.3(0.2) = \underline{247.34 \text{ volts}}$

$$\phi/\text{pole} = \frac{E_b \times 60A}{ZNP} = \frac{247.34 \times 60 \times 2}{220 \times 908 \times 6}$$

$$\boxed{\phi/\text{pole} = 0.02476 \text{ webers}}$$

P5] A 240V, 4 pole dc shunt motor gives (or) delivers 15 HP at a speed of 1000rpm, with armature current of 50A and a field current of 1A. The armature resistance is 0.1Ω and total contact drop is 2V. The armature is wave connected with 540 Conductors. Determine

(i) Shaft torque (or) net torque (or) Useful torque (T_{Sh})

(ii) Gross torque (or) Armature torque (or) developed torque (T_a)

(iii) useful flux per pole

(iv) Mechanical losses (or) Rotational losses

(v) Efficiency.

Data given

$V = 240$ Volts, $P = 4$, $P_{Sh} = 15 \times 735.5$

$I_a = 50A$, $I_f = 1A$ $P_{Sh} = 11032.5W$

$N = 1000$rpm, $R_a = 0.1\Omega$, $V_b = 2$Volts

$A = 2$(wave), $Z = 540$.

(i) $T_{Sh} = \frac{P_{Sh} \times 60}{2\pi N} = \frac{11032.5 \times 60}{2\pi(1000)} = 105.35$ N-m

(iii) $\phi_{/pole} = \frac{E_b \times 60A}{ZNP}$ where

$= \frac{233 \times 60 \times 2}{540 \times 1000 \times 4}$

$E_b = V - I_a R_a - V_b$

$= 240 - 50 \times 0.1 - 2$

$E_b = 233$ Volts

$\phi_{/pole} = 0.0129$ webers

(ii) $T_a = 0.159 \phi Z P \frac{I_a}{A} = 0.159 \times 0.0129 \times 540 \times 4 \times \frac{50}{2}$

$T_a = 111.249 N\text{-}m$

(iv) Mechanical losses & Rotational losses

$= P_d - P_{sh}$

$= E_b I_a - P_{sh}$

$= 233 \times 50 - 11032.5$

$= 617.5$ watts

(v) $\eta = \frac{P_{sh}}{V I_L} \times 100 = \frac{11032.5}{240(50+1)} \times 100 = 90.13\%$

where $I_L = I_a + I_f = 50 + 1 = 51A$

P6] A 4 Pole, 250 volts, dc series motor has wave connected armature with 1254 conductors. The flux/pole is 22mwb, when the motor is taking 50A. The armature and series field resistances are respectively 0.3Ω & 0.2Ω. Calculate (i) the speed (ii) Torque developed (iii) gross power.

Data given

$P = 4$, $V = 250$ volts, $A = 2$ (wave)

$Z = 1254$. $\phi/\text{pole} = 22 \times 10^{-3}$, $I_L = I_f = I_a = I = 50A$

$R_a = 0.3\Omega$ $R_{sc} = 0.2\Omega$

(i) $N = \frac{\{V - I(R_a + R_{sc})\} 60A}{\phi Z P}$

$= \frac{\{250 - 50(0.3+0.2)\} 60 \times 2}{22 \times 10^{-3} \times 1254 \times 4}$

$N = 244.67$ rpm

(ii) $T_a = 0.159 \phi Z P \frac{I_a}{A}$

$= 0.159 \times 22 \times 10^{-3} \times 1254 \times 4 \times 50/2$

$T_a = 438.649$ N-m

(iii) $P_d = E_b I_a = 225 \times 50 = 11250$ W.

Example PMC-4.1: Determine the total torque developed in a 250V, 4 pole DC shunt motor with lap winding accommodated in 60 slots, each containing 20 conductors. The armature current is 50 A and the flux per pole is 23 mWb.

Example PMC-4.2: A 4 pole, 220 V, lap connected DC shunt motor has 36 slots, each slot containing 16 conductors. It draws a current of 40 A from the supply. The field and armature resistances are 110 Ω and 0.1 Ω respectively. The motor develops an output power of 6 kW. The flux per pole is 40 mWb. Calculate the (a) speed (b) torque developed by the armature and (c) shaft torque.

Example PMC-4.7: A 250 V DC shunt motor takes 6 A line current on no load and runs at 1000 rpm. The resistance of field winding and armature are 250 Ω and 0.2 Ω respectively. If the full load line current is 26 A, calculate the full load speed, assuming constant air gap flux.

Example PMC-4.8: A 250 V DC shunt motor on no load runs at 1000 rpm and takes 6 A. The total armature and shunt field resistances are 0.2 Ω and 250 Ω respectively. Calculate the speed when loaded and taking a current of 50 A, if the armature reaction weakens the field by 3 %.

Example PMC-4.13: A 240 V, 4 pole DC shunt motor running at 1000 rpm gives 15 H.P. with an armature current of 50 A and a field current of 1.0 A. the armature winding is wave connected and has 540 conductors. Its resistance is 0.1 Ω and the drop at each brush is 1 V. Find (a) useful torque (b) total torque (c) useful flux per pole (d) rotational losses (e) efficiency.

Example BLT-29.1: A 220-V DC motor has an armature resistance of 0.5 Ω. If the full load armature current is 20 A, find the induced EMF. [**Ans:** 210 V]

Example BLT-29.7: A 250-V, 4-pole, wave wound DC series motor has 782 conductors on its armature. It has armature and series field resistance of 0.75 ohm. The motor takes a current of 40 A. Estimate its speed and gross torque developed if it has a flux per pole of 25 mWb. [**Ans:** 337.6 rpm, 249 Nm]

Example ELE-23: The armature of a 4-pole shunt motor is lap wound, which generates 216 V when running at 10 rps. It has 144 slots and 6 conductors per slot. (i) Determine the useful flux per pole (ii) If this armature is rewound as wave connected, determine the induced emf at the same speed and same flux per pole. [**Ans:** 0.025 Wb, 432 V]

Example ELE-24: A 4 pole lap connected DC motor has 600 armature conductors and runs at 1200 rpm. The motor has total flux of 0.24 Wb in it. Determine (i) induced emf (ii) speed at which it should be driven to produce the same emf when wave connected. [**Ans:** 720 V, 600 rpm]

Example ELE-32: A 500 V shunt motor has 4 poles and wave connected winding with 492 conductors. The flux per pole is 0.05 Wb. The full load current is 20 A. The armature and shunt field resistances are 0.1 Ω and 250 Ω respectively. Calculate the speed and gross torque. [**Ans:** 140.81 Nm, 607.56 rpm

Example ELE-33: A 120 V DC shunt motor has an armature resistance of 0.2 Ω and field resistance of 60. The full load line current is 60 A and full load speed is 1800 rpm. If brush contact drop is 3 V, find the speed of motor at half load. [**Ans:** 1902 rpm]

Example ELE-33: Find the useful flux per pole of a 250 V, 6 pole shunt motor having 2 circuit connected (or wave connected) armature winding with 220 conductors. The overall armature resistance including brushes is 0.2 Ω. The armature current is 13.3 A at the no load speed of 908 rpm. [**Ans:** 0.02476 Wb]

Example ELE-35: A 4 pole DC shunt motor takes 22 A from 200 V supply. The armature and field resistances are 0.5 Ω and 100 Ω respectively. The armature is lap connected with 300 conductors. If flux per pole is 20 mWb, find the speed and gross torque. [**Ans:** 1900 rpm, 19.08 Nm]

Example ELE-36: A DC series motor runs at 600 rpm and draws a current of 110 A from 250V supply. The armature resistance is 0.12 Ω and that of series field winding is 0.03 Ω. The useful flux per pole for 110 A is 0.024 Wb and that for 50 A is 0.0155 Wb. Calculate the speed when the current has fallen to 50 A. [**Ans:** 964.84 rpm]

Example ELE-37: A 440 V, DC shunt motor takes an armature current of 20 A and runs at 500 rpm. The armature resistance is 0.6 Ω. If the flux is reduced by 30% and the torque is increased by 40%, what are the new values of armature current and the speed? [**Ans:** 40 A, 694.26 rpm]

Example ELE-38: A 240V, 4 pole shunt motor running at 1000 rpm delivers 15 HP with an armature current of 50 A and a field current of 1 A. The armature winding is wave connected and has 540 conductors. Its resistance is 0.1 Ω and drop at each brush is 1 V. Find (i) useful torque or net torque or shaft torque (ii) gross torque or armature torque (iii) useful flux per pole (iv) rotational losses or mechanical losses and (v) efficiency. [**Ans:** 105.35 Nm, 110.76 Nm, 0.0129 Wb, 617.5 W, 90.134 %]

Example ELE-40: A 4 pole, 220 V lap connected DC shunt motor has 36 slots and 16 conductors per slot. It draws a current of 4 A from the supply. The field and armature resistances are 110 Ω and 0.1 Ω respectively. The motor delivers an output power of 6 kW, when the flux per pole is 40 mWb. Calculate the (i) speed (ii) torque developed by the armature and (iii) shaft torque. [**Ans:** 563 rpm, 139.34 Nm, 101.82 Nm]

Example ELE-41: A 4 pole 250 V series motor has wave connected armature with 1254 conductors. The flux per pole is 22 mWb, when the motor is taking 50 A. the armature and series field coil resistances are 0.3 Ω and 0.2 Ω respectively. Calculate (i) speed (ii) torque of the motor and (iii) power developed in kW. [**Ans:** 244.67 rpm, 438.65 Nm, 11.25 kW]